UJJAIN ENGINEERING COLLEGE, UJJAIN

Indore Road, Ujjain Madhya Pradesh, 456010

Department of Computer Science and Engineering

Name of the Subject & Code : Internet of Things (CS-6411)
Name of the faculty Member : Asst. Prof. Rekha Singh
Semester : VI Semester

S.No.	Experiments	Date of Performance	Date of Submission	Remarks
1.	Install Arduino IDE and Blink an LED to verify setup.			
2.	Interface a temperature and humidity sensor (e.g., LM35 or DHT11) with Arduino / Raspberry Pi and display readings on serial monitor.			
3.	Interface a relay or motor and control it via GPIO pins.			
4.	Interface RFID reader (RC522) with Arduino/Raspberry Pi.Read RFID tag ID and display it on the serial monitor or LCD.			
5.	Establish Bluetooth communication between two devices and Send sensor data (temperature) to mobile app via Bluetooth.			
6.	Connect an IoT device (NodeMCU/ESP8266) to a Wi-Fi network and upload sensor data to the cloud (ThingSpeak) for remote monitoring.			
7.	Read data from humidity and light sensors simultaneously and log values.			
8.	Control brightness of an LED or speed of a DC motor using PWM on Arduino.			
9.	Combine temperature, humidity, and light sensors to create a simple weather station.			
10.	Send sensor data from one microcontroller to another via UART/I2C/SPI.			
11.	Publish and subscribe to topics using an MQTT broker (Mosquitto) by sending temperature data to the broker and controlling an LED from the broker.			
12.	To establish secure communication in an IoT ecosystem by enabling HTTPS/TLS while sending data to a cloud platform, and to transmit sensor data from an IoT device using both TCP and UDP protocols.			
13.	Mini Project (Automatic Fire Extinguisher System).			